



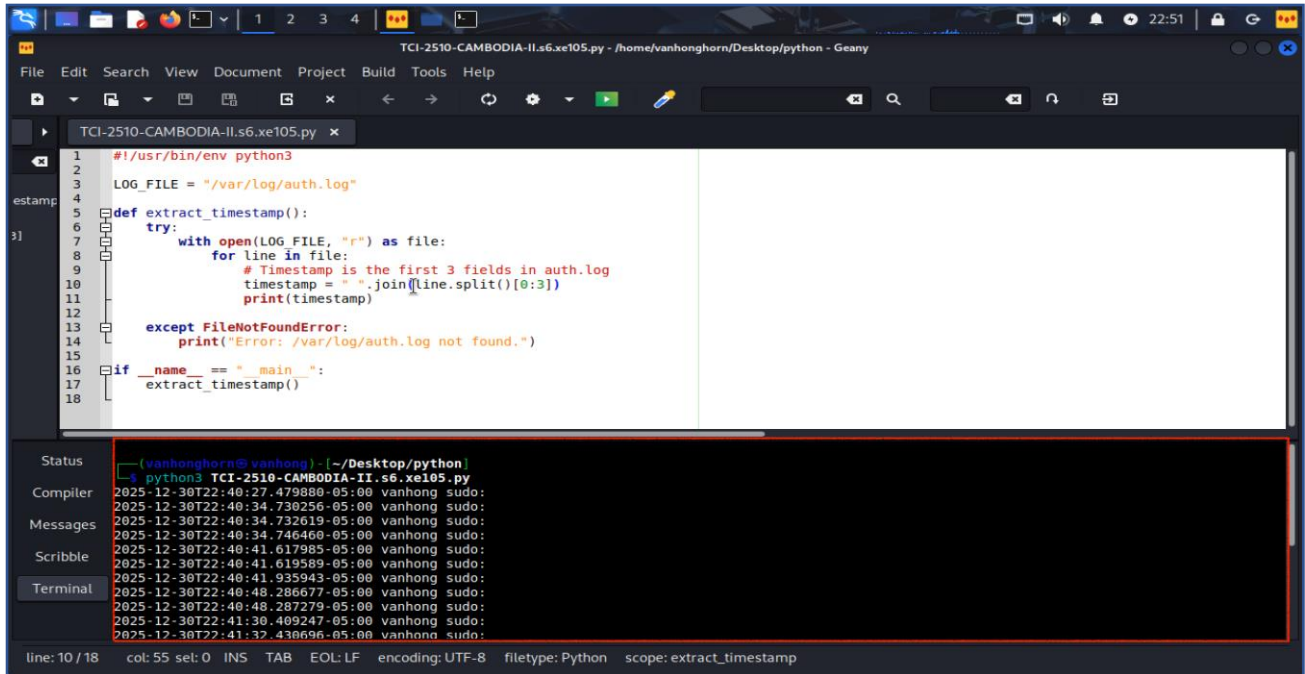
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LOG ANALYZER

1. Log Parse auth.log: Extract command usage.

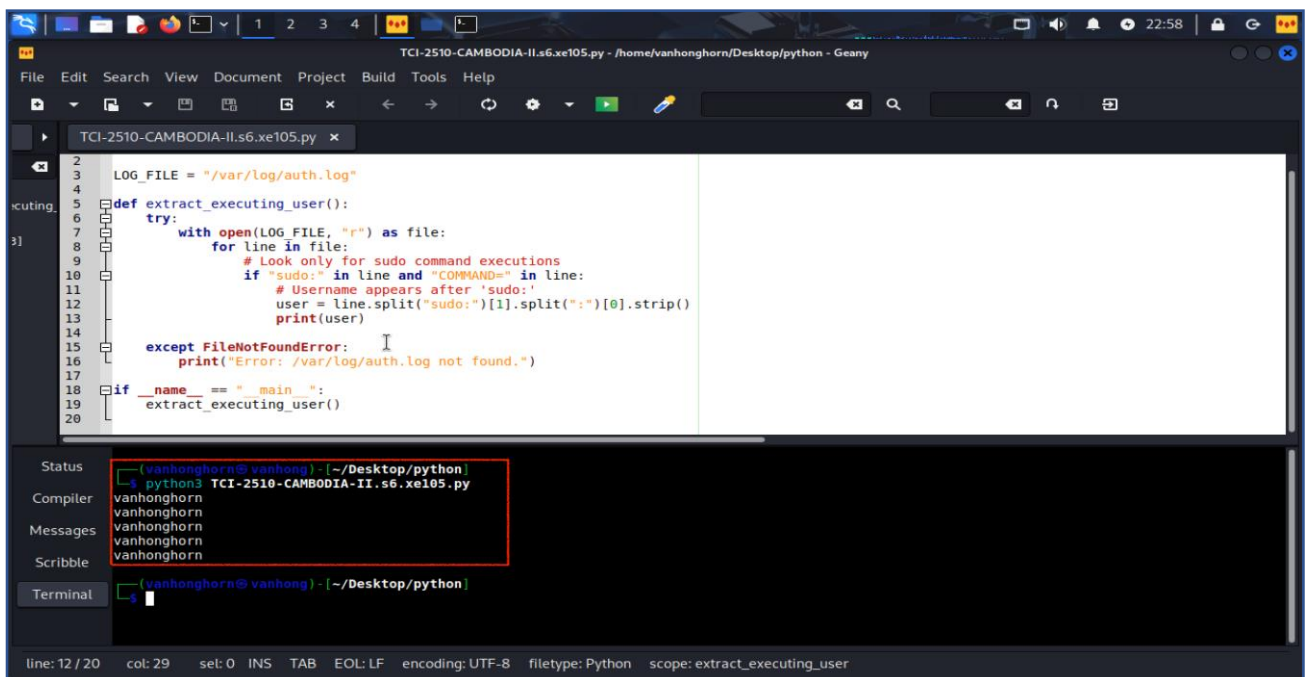
1.1. Include the Timestamp.



```
TCI-2510-CAMBODIA-II.s6.xe105.py - /home/vanhonghorn/Desktop/python - Geany
File Edit Search View Document Project Build Tools Help
TCI-2510-CAMBODIA-II.s6.xe105.py x
1 #!/usr/bin/env python3
2
3 LOG_FILE = "/var/log/auth.log"
4
5
6 def extract_timestamp():
7     try:
8         with open(LOG_FILE, "r") as file:
9             for line in file:
10                 # Timestamp is the first 3 fields in auth.log
11                 timestamp = " ".join(line.split()[0:3])
12                 print(timestamp)
13
14     except FileNotFoundError:
15         print("Error: /var/log/auth.log not found.")
16
17 if __name__ == "__main__":
18     extract_timestamp()
```

Status (vanhonghorn@vanhong) - [~/Desktop/python]
Compiler python3 TCI-2510-CAMBODIA-II.s6.xe105.py
Messages 2025-12-30T22:40:27.479880-05:00 vanhong sudo:
2025-12-30T22:40:34.730256-05:00 vanhong sudo:
2025-12-30T22:40:34.732619-05:00 vanhong sudo:
2025-12-30T22:40:34.746460-05:00 vanhong sudo:
2025-12-30T22:40:41.617905-05:00 vanhong sudo:
2025-12-30T22:40:41.619589-05:00 vanhong sudo:
2025-12-30T22:40:41.935943-05:00 vanhong sudo:
2025-12-30T22:40:48.286677-05:00 vanhong sudo:
2025-12-30T22:40:48.287279-05:00 vanhong sudo:
2025-12-30T22:41:30.409247-05:00 vanhong sudo:
2025-12-30T22:41:32.430696-05:00 vanhong sudo:
line: 10/18 col: 55 set: 0 INS TAB EOL: LF encoding: UTF-8 filetype: Python scope: extract_timestamp

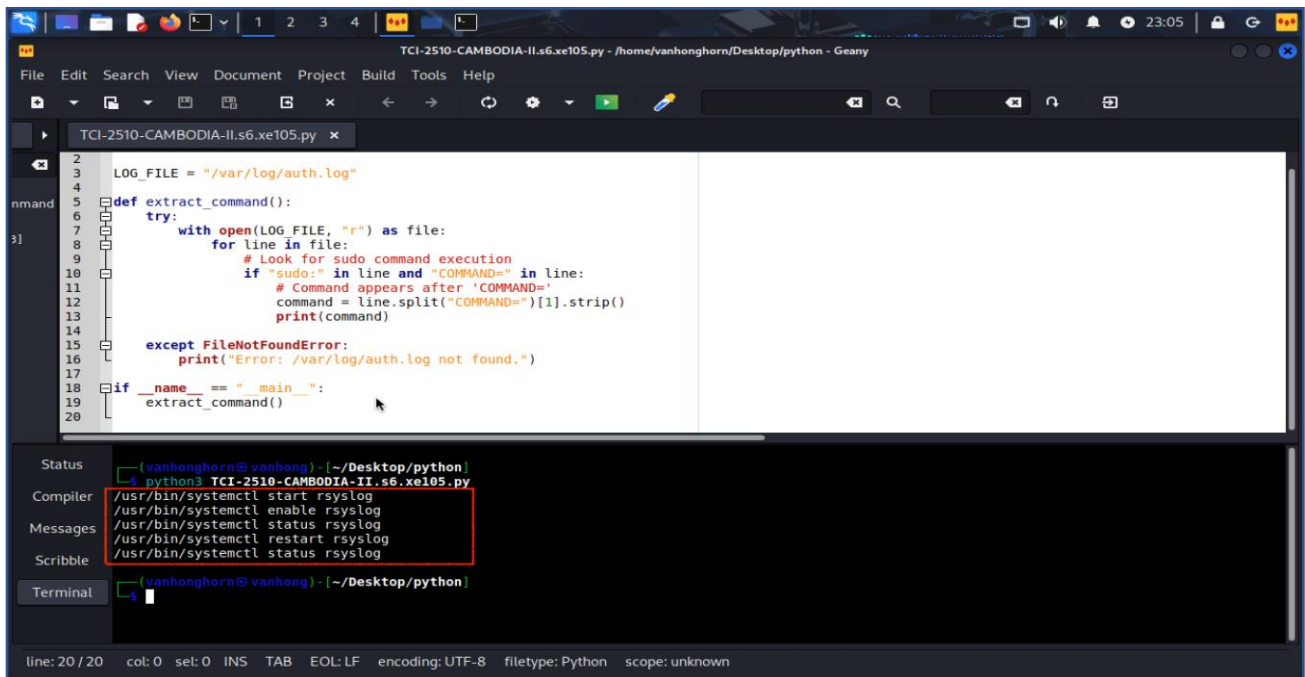
1.2. Include the executing user.



```
TCI-2510-CAMBODIA-II.s6.xe105.py - /home/vanhonghorn/Desktop/python - Geany
File Edit Search View Document Project Build Tools Help
TCI-2510-CAMBODIA-II.s6.xe105.py x
2 LOG_FILE = "/var/log/auth.log"
3
4
5 def extract_executing_user():
6     try:
7         with open(LOG_FILE, "r") as file:
8             for line in file:
9                 # Look only for sudo command executions
10                 if "sudo:" in line and "COMMAND=" in line:
11                     # Username appears after 'sudo:'
12                     user = line.split("sudo:")[1].split(":")[0].strip()
13                     print(user)
14
15     except FileNotFoundError:
16         print("Error: /var/log/auth.log not found.")
17
18 if __name__ == "__main__":
19     extract_executing_user()
```

Status (vanhonghorn@vanhong) - [~/Desktop/python]
Compiler python3 TCI-2510-CAMBODIA-II.s6.xe105.py
Messages vanhonghorn
vanhonghorn
vanhonghorn
vanhonghorn
vanhonghorn
Terminal (vanhonghorn@vanhong) - [~/Desktop/python]
line: 12/20 col: 29 set: 0 INS TAB EOL: LF encoding: UTF-8 filetype: Python scope: extract_executing_user

1.3. Include the command.



The screenshot shows a code editor with a Python script named `TCI-2510-CAMBODIA-II.s6.xe105.py`. The script defines a function `extract_command()` that reads from `/var/log/auth.log` and prints the command part of lines containing `sudo`. The terminal output shows the command `python3 TCI-2510-CAMBODIA-II.s6.xe105.py` being executed, which results in several `systemctl` commands being printed.

```
LOG_FILE = "/var/log/auth.log"

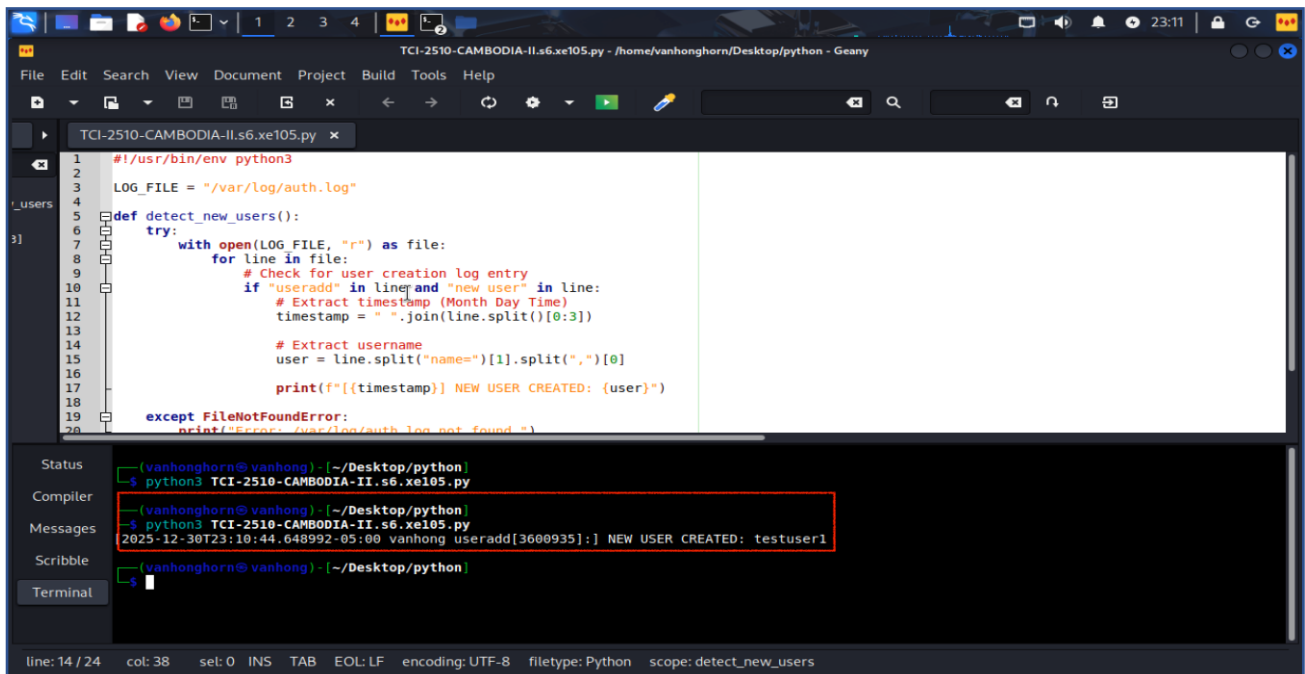
def extract_command():
    try:
        with open(LOG_FILE, "r") as file:
            for line in file:
                # Look for sudo command execution
                if "sudo:" in line and "COMMAND=" in line:
                    # Command appears after 'COMMAND='
                    command = line.split("COMMAND=")[1].strip()
                    print(command)
    except FileNotFoundError:
        print("Error: /var/log/auth.log not found.")

if __name__ == "__main__":
    extract_command()
```

```
(vanhonghorn@ vanhong) ~/Desktop/python
$ python3 TCI-2510-CAMBODIA-II.s6.xe105.py
/usr/bin/systemctl start rsyslog
/usr/bin/systemctl enable rsyslog
/usr/bin/systemctl status rsyslog
/usr/bin/systemctl restart rsyslog
/usr/bin/systemctl status rsyslog
```

2. Log Parse auth.log: Monitor user authentication changes.

2.1. Print details of newly added users, including the Timestamp.



The screenshot shows a code editor with a Python script named `TCI-2510-CAMBODIA-II.s6.xe105.py`. The script defines a function `detect_new_users()` that reads from `/var/log/auth.log` and prints the timestamp and username of newly created users. The terminal output shows the command `python3 TCI-2510-CAMBODIA-II.s6.xe105.py` being executed, which results in a message being printed: `2025-12-30T23:10:44.648992-05:00 vanhong useradd[3600935:] NEW USER CREATED: testuser1`.

```
#!/usr/bin/env python3
LOG_FILE = "/var/log/auth.log"

def detect_new_users():
    try:
        with open(LOG_FILE, "r") as file:
            for line in file:
                # Check for user creation log entry
                if "useradd" in line and "new user" in line:
                    # Extract timestamp (Month Day Time)
                    timestamp = " ".join(line.split()[0:3])

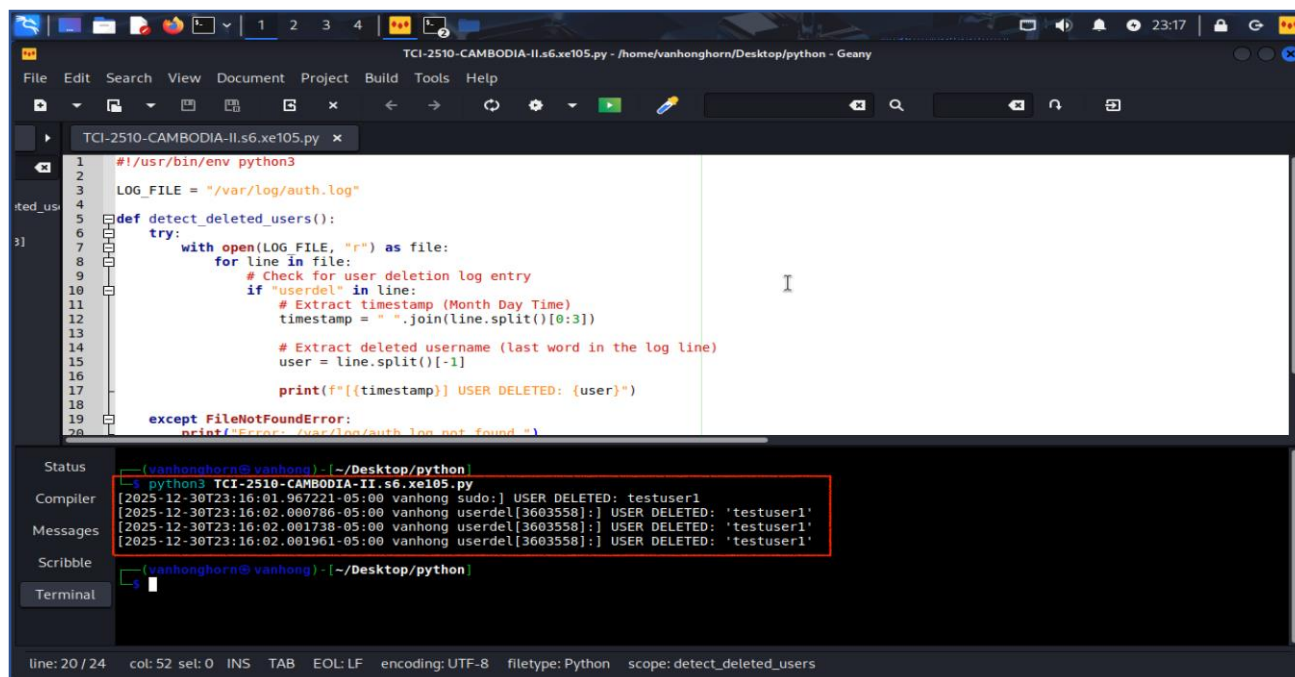
                    # Extract username
                    user = line.split("name=")[1].split(",")[0]

                    print(f"[{timestamp}] NEW USER CREATED: {user}")
    except FileNotFoundError:
        print("Error: /var/log/auth.log not found.")

if __name__ == "__main__":
    detect_new_users()
```

```
(vanhonghorn@ vanhong) ~/Desktop/python
$ python3 TCI-2510-CAMBODIA-II.s6.xe105.py
2025-12-30T23:10:44.648992-05:00 vanhong useradd[3600935:] NEW USER CREATED: testuser1
```

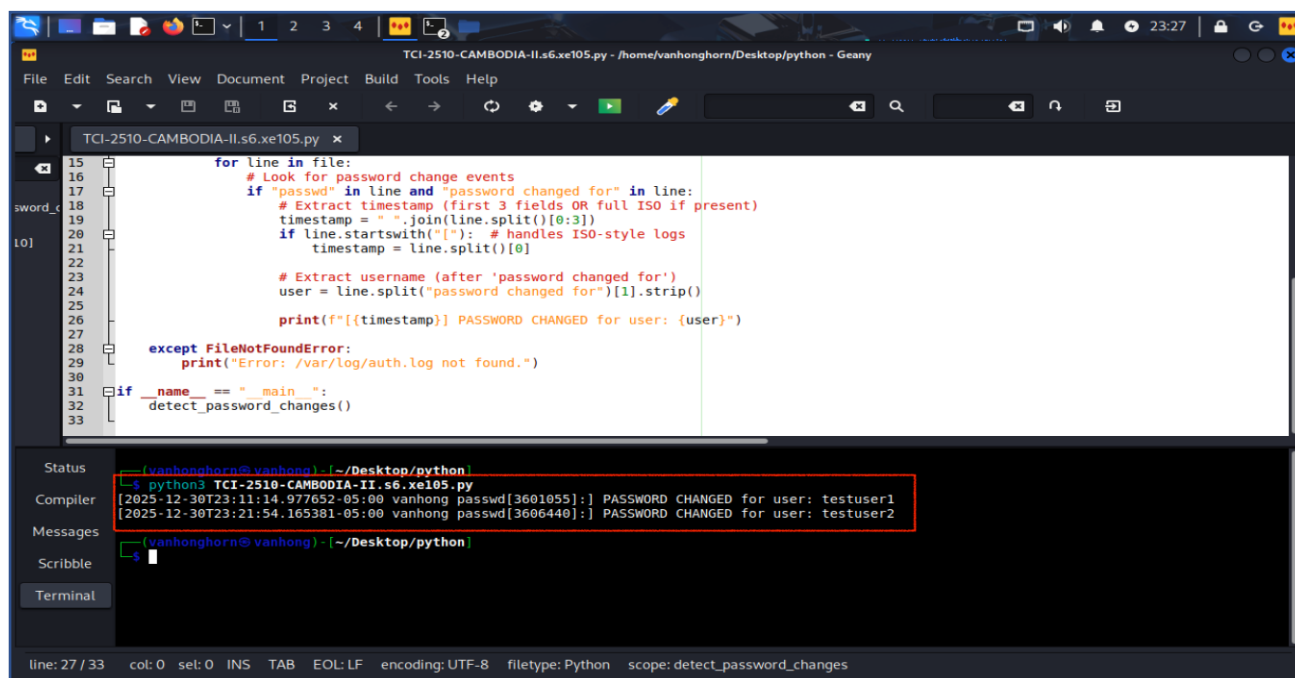
2.2. Print details of deleted users, including the Timestamp.



The screenshot shows the Geany IDE with a Python script named `TCI-2510-CAMBODIA-II.s6.xe105.py`. The script defines a function `detect_deleted_users()` that reads the `/var/log/auth.log` file and prints the timestamp and username of deleted users. The terminal output shows the script being executed and the following messages:

```
(vanhonghorn@ vanhong) - [~/Desktop/python]
$ python3 TCI-2510-CAMBODIA-II.s6.xe105.py
[2025-12-30T23:16:01.967221-05:00 vanhong sudo:] USER DELETED: testuser1
[2025-12-30T23:16:02.000786-05:00 vanhong userdel[3603558]:] USER DELETED: 'testuser1'
[2025-12-30T23:16:02.001738-05:00 vanhong userdel[3603558]:] USER DELETED: 'testuser1'
[2025-12-30T23:16:02.001961-05:00 vanhong userdel[3603558]:] USER DELETED: 'testuser1'
```

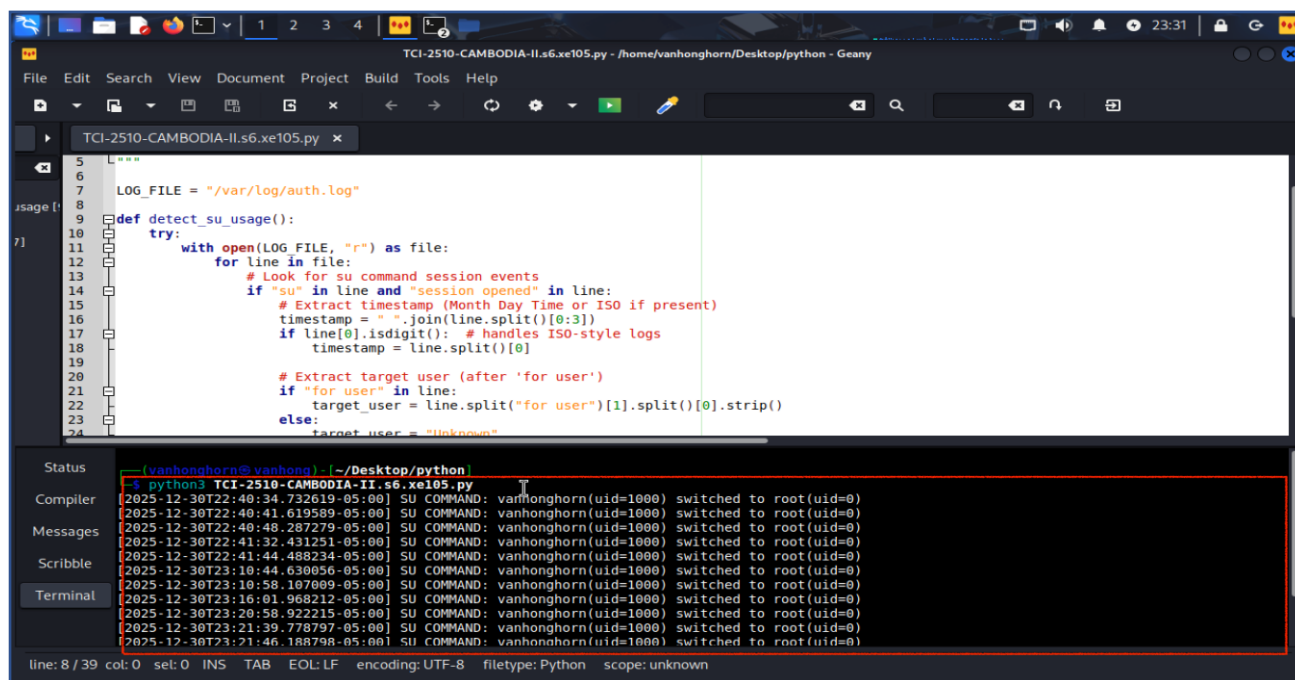
2.3. Print details of changing passwords, including the Timestamp.



The screenshot shows the Geany IDE with a Python script named `TCI-2510-CAMBODIA-II.s6.xe105.py`. The script defines a function `detect_password_changes()` that reads the `/var/log/auth.log` file and prints the timestamp and username of password changes. The terminal output shows the script being executed and the following messages:

```
(vanhonghorn@ vanhong) - [~/Desktop/python]
$ python3 TCI-2510-CAMBODIA-II.s6.xe105.py
[2025-12-30T23:11:14.977652-05:00 vanhong passwd[3601055]:] PASSWORD CHANGED for user: testuser1
[2025-12-30T23:21:54.165381-05:00 vanhong passwd[3606440]:] PASSWORD CHANGED for user: testuser2
```


2.4. Print details of when users used the su command.



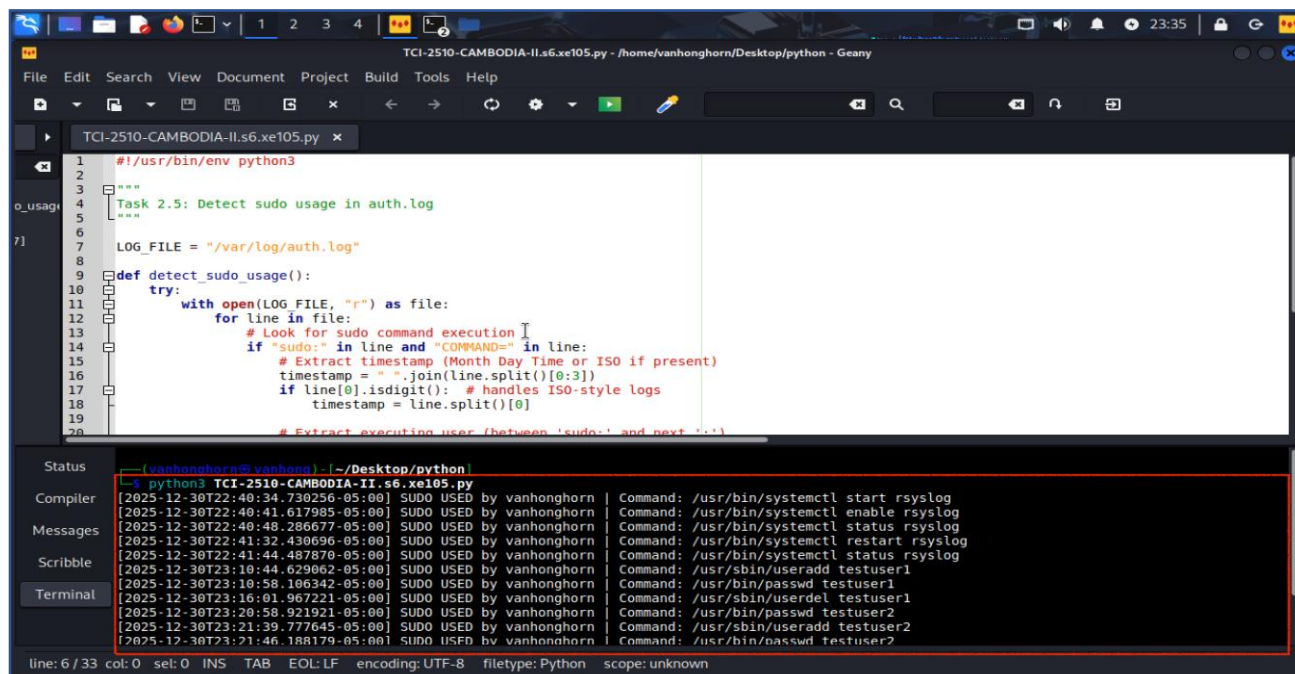
The screenshot shows a Geany IDE window titled "TCI-2510-CAMBODIA-II.s6.xe105.py - /home/vanhonghorn/Desktop/python - Geany". The editor displays a Python script named `detect_su_usage.py` with the following code:

```
5 """
6
7 LOG_FILE = "/var/log/auth.log"
8
9 def detect_su_usage():
10     try:
11         with open(LOG_FILE, "r") as file:
12             for line in file:
13                 # Look for su command session events
14                 if "su" in line and "session opened" in line:
15                     # Extract timestamp (Month Day Time or ISO if present)
16                     timestamp = " ".join(line.split()[0:3])
17                     if line[0].isdigit(): # handles ISO-style logs
18                         timestamp = line.split()[0]
19
20                     # Extract target user (after 'for user')
21                     if "for user" in line:
22                         target_user = line.split("for user")[1].split()[0].strip()
23                     else:
24                         target_user = "Unknown"
```

The terminal output shows the execution of the script, displaying a list of su command sessions with timestamps and target users:

```
(vanhonghorn@vanhong) ~/Desktop/python
python3 TCI-2510-CAMBODIA-II.s6.xe105.py
[2025-12-30T22:40:34.732619-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T22:40:41.619589-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T22:40:48.287279-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T22:41:32.431251-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T22:41:44.488234-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T23:10:44.630656-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T23:10:58.107009-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T23:16:01.968212-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T23:20:58.922215-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T23:21:39.778797-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
[2025-12-30T23:21:46.188798-05:00] SU COMMAND: vanhonghorn(uid=1000) switched to root(uid=0)
```

2.5. Print details of users who used the sudo; include the command.



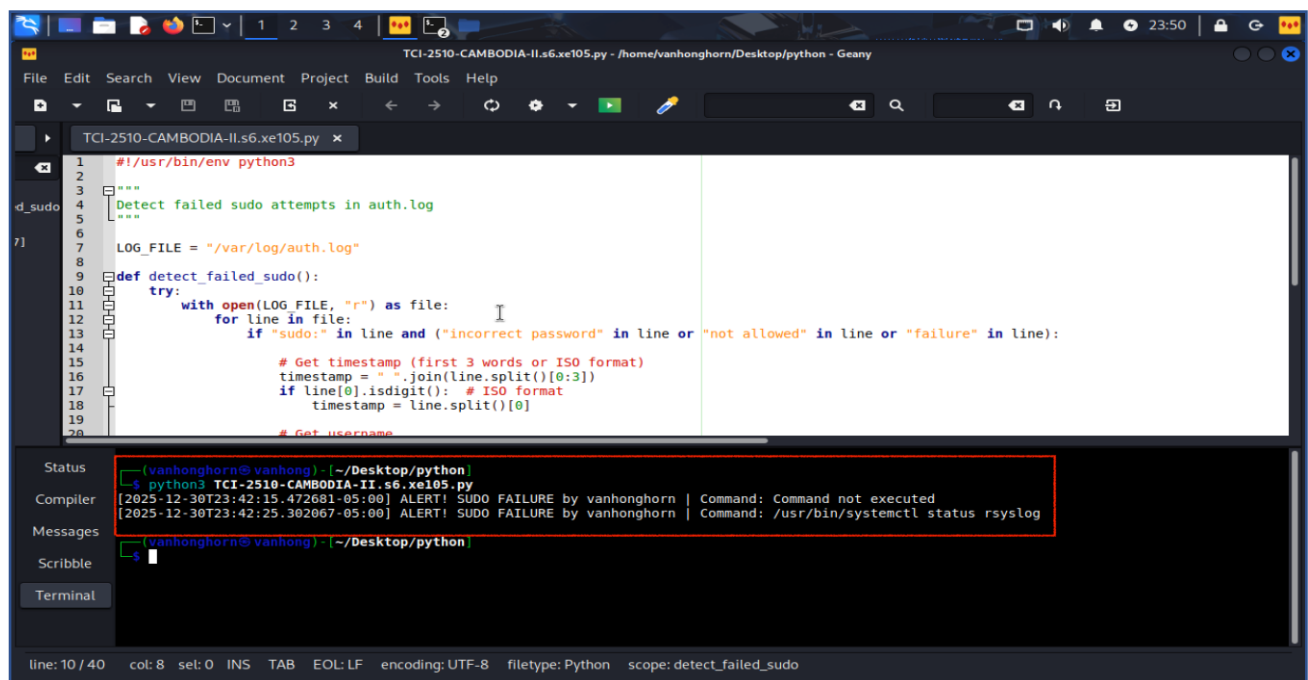
The screenshot shows a Geany IDE window titled "TCI-2510-CAMBODIA-II.s6.xe105.py - /home/vanhonghorn/Desktop/python - Geany". The editor displays a Python script named `detect_sudo_usage.py` with the following code:

```
1 #!/usr/bin/env python3
2
3 """
4 Task 2.5: Detect sudo usage in auth.log
5 """
6
7 LOG_FILE = "/var/log/auth.log"
8
9 def detect_sudo_usage():
10     try:
11         with open(LOG_FILE, "r") as file:
12             for line in file:
13                 # Look for sudo command execution
14                 if "sudo:" in line and "COMMAND=" in line:
15                     # Extract timestamp (Month Day Time or ISO if present)
16                     timestamp = " ".join(line.split()[0:3])
17                     if line[0].isdigit(): # handles ISO-style logs
18                         timestamp = line.split()[0]
19
20                     # Extract executing user (between 'sudo:' and next ' ')
21                     user = line.split("sudo:")[1].split()[0]
```

The terminal output shows the execution of the script, displaying a list of sudo command executions with timestamps, users, and commands:

```
(vanhonghorn@vanhong) ~/Desktop/python
python3 TCI-2510-CAMBODIA-II.s6.xe105.py
[2025-12-30T22:40:34.730256-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/systemctl start rsyslog
[2025-12-30T22:40:41.617985-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/systemctl enable rsyslog
[2025-12-30T22:40:48.286677-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/systemctl status rsyslog
[2025-12-30T22:41:32.430696-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/systemctl restart rsyslog
[2025-12-30T22:41:44.487870-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/systemctl status rsyslog
[2025-12-30T23:10:44.629062-05:00] SUDO USED by vanhonghorn | Command: /usr/sbin/useradd testuser1
[2025-12-30T23:10:58.106342-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/passwd testuser1
[2025-12-30T23:16:01.967221-05:00] SUDO USED by vanhonghorn | Command: /usr/sbin/userdel testuser1
[2025-12-30T23:20:58.921921-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/passwd testuser2
[2025-12-30T23:21:39.777645-05:00] SUDO USED by vanhonghorn | Command: /usr/sbin/useradd testuser2
[2025-12-30T23:21:46.188179-05:00] SUDO USED by vanhonghorn | Command: /usr/bin/passwd testuser2
```

2.6. Print ALERT! If users failed to use the sudo command; include the command.



The screenshot shows the Geany IDE with a Python script named `TCI-2510-CAMBODIA-II.s6.xe105.py` open. The script is designed to detect failed sudo attempts by monitoring the `/var/log/auth.log` file. It defines a function `detect_failed_sudo()` that reads the log file and checks for lines containing "sudo:" followed by "incorrect password", "not allowed", or "failure". When such a line is found, it extracts the timestamp and username, and prints an alert message.

```
1 #!/usr/bin/env python3
2
3 """
4 Detect failed sudo attempts in auth.log
5 """
6
7 LOG_FILE = "/var/log/auth.log"
8
9 def detect_failed_sudo():
10     try:
11         with open(LOG_FILE, "r") as file:
12             for line in file:
13                 if "sudo:" in line and ("incorrect password" in line or "not allowed" in line or "failure" in line):
14                     # Get timestamp (first 3 words or ISO format)
15                     timestamp = " ".join(line.split()[0:3])
16                     if line[0].isdigit(): # ISO format
17                         timestamp = line.split()[0]
18                     # Get username
19
20
```

The terminal output shows the script being executed and the resulting alert messages:

```
(vanhonghorn@ vanhong) - [~/Desktop/python]
$ python3 TCI-2510-CAMBODIA-II.s6.xe105.py
[2025-12-30T23:42:15.472681-05:00] ALERT! SUDO FAILURE by vanhonghorn | Command: Command not executed
[2025-12-30T23:42:25.302067-05:00] ALERT! SUDO FAILURE by vanhonghorn | Command: /usr/bin/systemctl status rsyslog
```

The status bar at the bottom indicates the current line is 10 of 40, column 8, and the file type is Python.